- 1. 随机变量$X \sim \mathcal{P}(3), Y \sim \mathcal{P}(6)$都是泊松分布，且X与Y独立. 请用卷积法求$Z = X + Y$的分布列、并写出对应分布名称与参数.

- 2. 随机变量$X \sim \Gamma(\alpha, \lambda), Y \sim \Gamma(\beta, \lambda)$，且X与Y独立. 请证明$Z = X + Y \sim \Gamma(\alpha + \beta, \lambda)$. (Hint: 化简时或可利用Beta分布)

Problem 14. Competing exponentials. The lifetimes of two light bulbs are modeled as independent and exponential random variables X and Y, with parameters λ and μ, respectively. The time at which a light bulb first burns out is

$$Z = \min\{X.Y\}.$$

Show that Z is an exponential random variable with parameter $\lambda + \mu$.

4. 设随机变量X服从参数为1的指数分布，随机变量$Y \sim N(0,1)$，且X与Y独立. 试求随机变量$Z = \sqrt{2X}\,|\,Y\,|$ 的概率密度函数.

- 5. 设有2个独立的连续型随机变量X_1, X_2，$X_1 \sim \mathcal{U}(1,2)$是均匀分布，$X_2$的概率密度为 $f_{X_2}(x_2) = \begin{cases} \frac{3}{8}x_2^2, & \text{if } x_2 \in [0,2] \\ 0, & \text{otherwise} \end{cases}$. 令$\begin{cases} Y_1 = X_1^2 \\ Y_2 = X_1 + X_2 \end{cases}$. 求$Y = (Y_1, Y_2)$的联合概率密度.

- 6.【知识点: 随机变(向)量的函数的分布及独立性(PDF and Independence of $g(X)$ or $g(\boldsymbol{X})$)，卡方分布、F分布】设有3个独立的随机变量X_1, X_2, X_3，且$X_i \sim \chi^2_{r_i}$, $i = 1,2,3$.
  - a) 设$Y_1 = \frac{X_1}{X_2}, Y_2 = X_1 + X_2$. 请证明$Y_1, Y_2$相互独立，且$Y_2 \sim \chi^2_{r_1+r_2}$.
  - b) 请说明$\frac{X_1/r_1}{X_2/r_2}$和$\frac{X_3/r_3}{(X_1+X_2)/(r_1+r_2)}$都服从F分布(注明具体参数)、且相互独立.

- 次序统计量
  - 7. 设$X_1,\ldots,X_n$独立同分布，X_1的概率密度为$f(x)$. 对于给定的常数整数$r \le n$，求$(X_{(1)},X_{(2)}\ldots,X_{(r)})$的联合概率密度.